

Project Development Phase

Dataset Description & Data Dictionary

Project Title	India's Agricultural Crop Production Analysis
Domain	Agriculture / Data Analytics & Business Intelligence
Tool	Tableau Desktop / Tableau Public
Dataset	Historical Indian Crop Production Data, 1997–2021
Prepared By	Divya Kauluri (Team Project — completed individually; the team lead and other assigned members did not contribute to the work)
Mentor	N/A — Individual Submission
GitHub Repository	https://github.com/divyakauluri-spec/India-s-agricultural-crop-production-analysis
Tableau Public Link	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject
Date	02/07/2026

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and

what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

Field	Value
Dataset Name	skillwallet_crop_cleaned.csv (cleaned extract)
Source	Historical Indian Crop Production Data, 1997-2021 (government agricultural statistics)
Record Count	345,407 rows
Coverage	36 states/union territories · 57 crop types · 6 growing seasons · 25 years (1997-2021)

1. Data Dictionary

Field	Type	Description
Country	Text	Fixed value 'India'; retained for potential future multi-country comparison
State_Name	Text (Dimension)	Name of the Indian state or union territory where the crop was recorded
District_Name	Text (Dimension)	District-level granularity within each state (available for future drill-down)
Crop_Year / Crop_Year_Numeric	Text / Number	Agricultural year of the record; the numeric version drives all trend charts
Season	Text (Dimension)	Growing season — Kharif, Rabi, Autumn, Summer, Winter, or Whole Year
Crop	Text (Dimension)	Crop name (e.g., Rice, Wheat, Cotton(lint), Soyabean, Bajra, Maize, Jowar, Gram, Groundnut, Rapeseed & Mustard)
Area / Area (M)	Number	Cultivated area for the record, in original units and in millions (M)
Area_Units	Text	Unit of the Area field (Hectare)
Production / Production (M)	Number	Total quantity produced, in original units and in millions (M)
Production_Units	Text (Dimension)	Unit of measure for Production — Tonnes, Bales, or Nuts — differ by crop
Production_Normalized	Number (Calculated)	Production converted to a common tonnage-equivalent basis where feasible
Yield	Number (Calculated)	Production divided by Area, representing productivity per unit of land

2. Cleaning & Preparation Summary

Duplicate and blank rows were removed, state and crop names were standardised, Crop_Year_Numeric and Yield were derived, Production_Normalized was calculated to account for mixed units, and Area/Production were scaled to

millions for chart readability. Full step-by-step detail is documented in the Preprocessing Steps & Business Questions file in this same folder.

3. Known Limitations

- 2021 figures are partial-year, since year-end government reporting was incomplete at the time of data collection.
- Production_Units differ by crop (Tonnes, Bales, Nuts); comparisons across crops or states should always be read alongside the Production Distribution by Unit Type chart.
- District-level data exists in the schema but was not used for district-level charts in this phase of the project; flagged as a Future Scope item.